

Week 4
Parallel Robots

ENN581 Robot Motion, Control and Planning

Dr. Krishna Manaswi Digumarti

School of Electrical Engineering and Robotics

ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Last week, we learnt

- Inverse kinematics is the problem of determining how to get the robot to a desired pose.
- For a serial robot, an analytical solution exists sometimes and can be arrived at by using algebra or geometry.
- When an analytical solution doesn’t exist, a numerical method can be used.
- All systems with revolute and prismatic joints having a total of 6 DoF in a single series chain are solvable.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Feedback on assessment 2.1

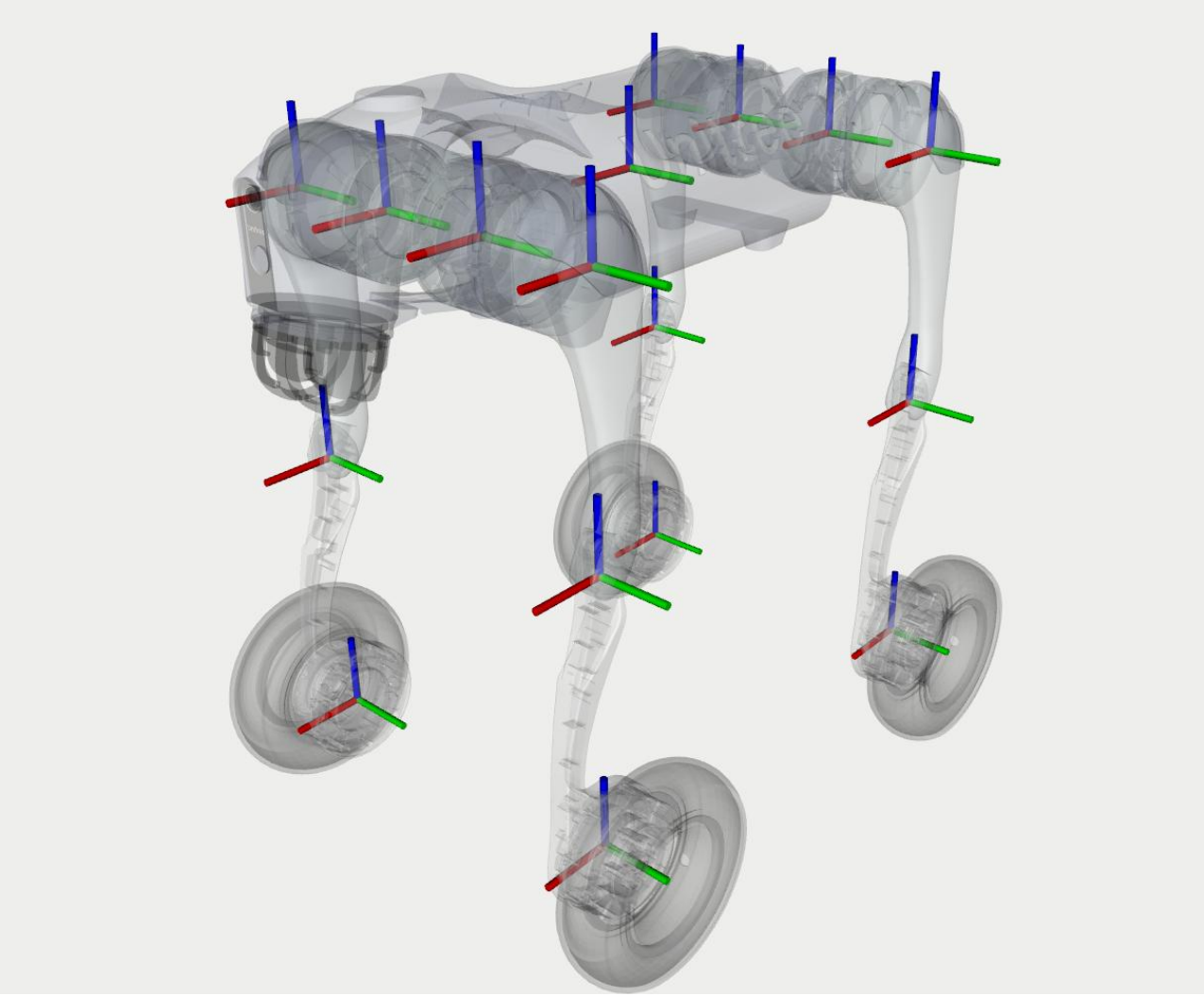
- Keep things simple. No need to do complex array manipulation.
- Write down on a piece of paper, what the question is asking you to do and what each parameter means. Write pseudocode before you write code.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Unitree As2-W Powerful Pioneer, Compact Yet Extraordinary | Unitree Robotics



Unitree Go2



The rotation axis of the fuselage joint is the x-axis, the rotation axis of the thigh joint and the calf joint is the y-axis, and the positive dire ction of rotation conforms to the right-hand rule.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Your task in assessment 2.2



Today's lecture

- Parallel robots
  - Various constructions of parallel chain mechanisms
  - Comparison with serial chain mechanisms
- Mobility of parallel robots
- Workspace of parallel robots
- Forward kinematics of parallel robots
- Inverse kinematics of parallel robots

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Serial chain manipulators are great

- Large working volume and high dexterity,

but

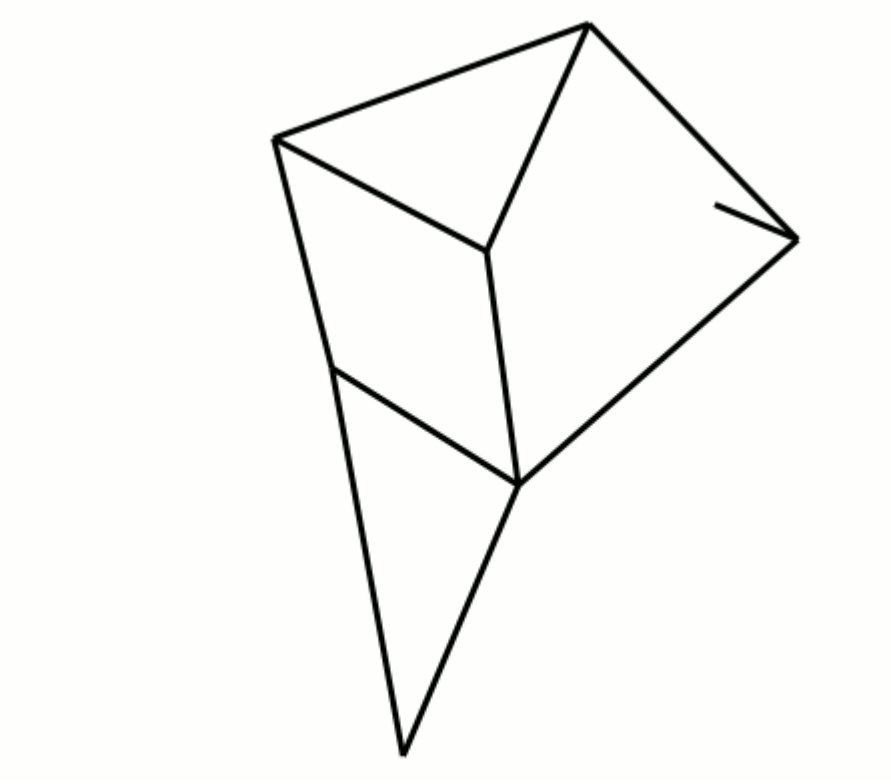
- are not very precise
- have poor force exertion capability
- have motors distributed throughout the chain
- have many moving parts – high inertia

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J



Theo Jansen's walkers, BBC One

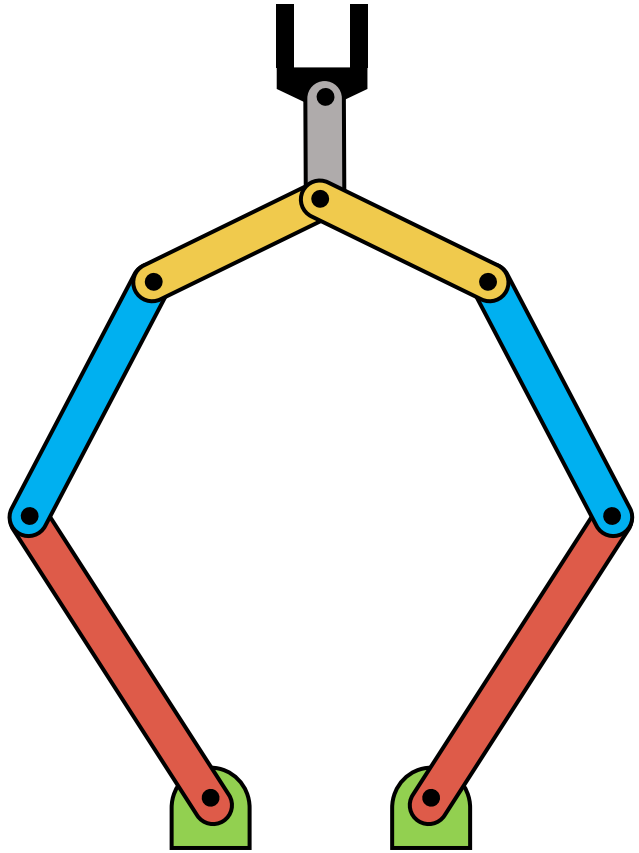
Many joints but only 1 degree of freedom



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

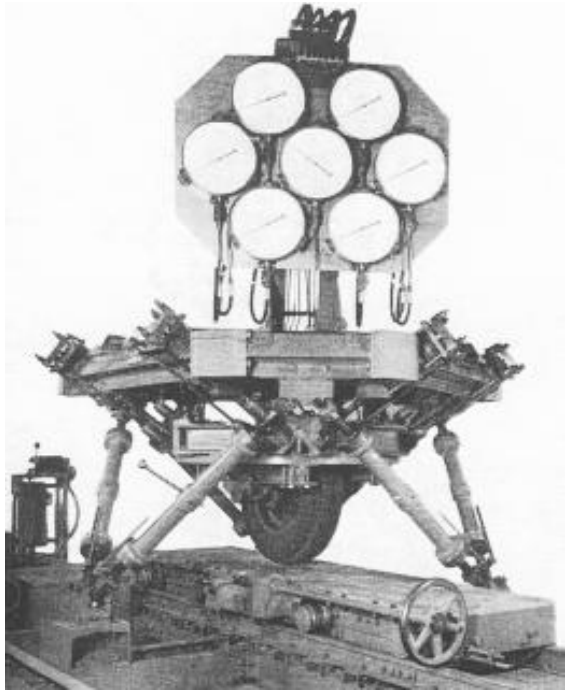
Parallel mechanism

- The end effector is connected to the base by multiple parallel kinematic chains of links.
- The arrangement can be both planar and spatial.
- Each actuator may have a wide range of motion, but the motion of the end effector might be constrained
- The number of joints is usually more than the degrees of freedom of the mechanism – several joints are passive



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

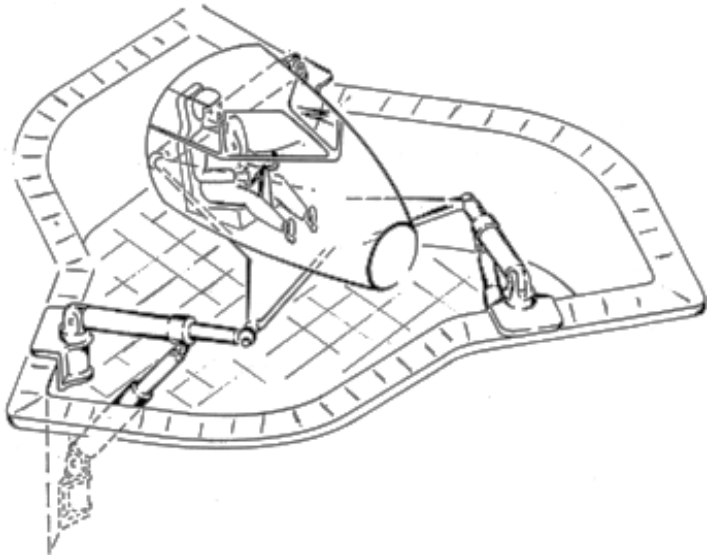
One of the early parallel robots was built to test tires



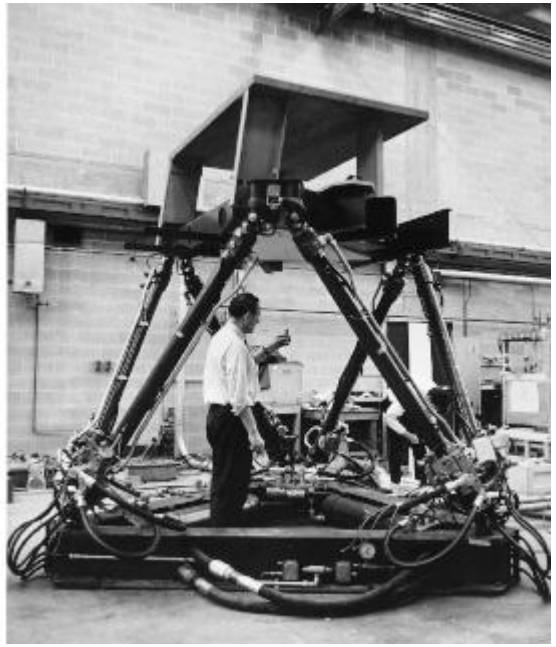


Universal Tyre-Testing Machine invented by Eric Gough at Dunlop, early 1950s and now

Parallel robots became popular for flight simulators



Stewart's flight simulator, 1965



Klaus Cappel's flight simulator, mid 1960s



A car simulator

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

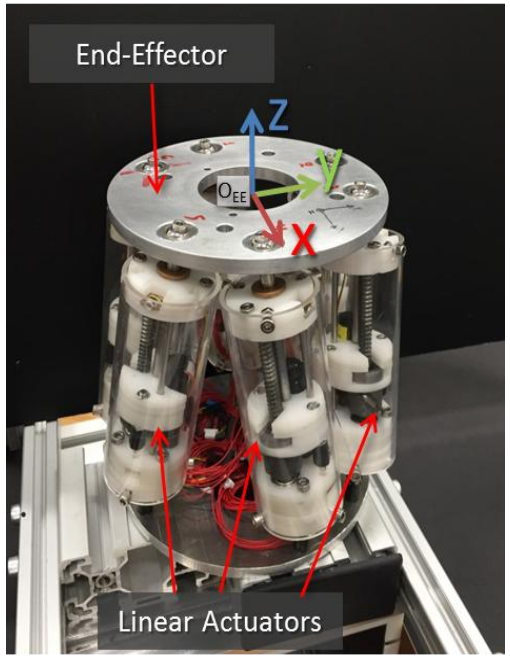
Now they are widely used in pick and place tasks



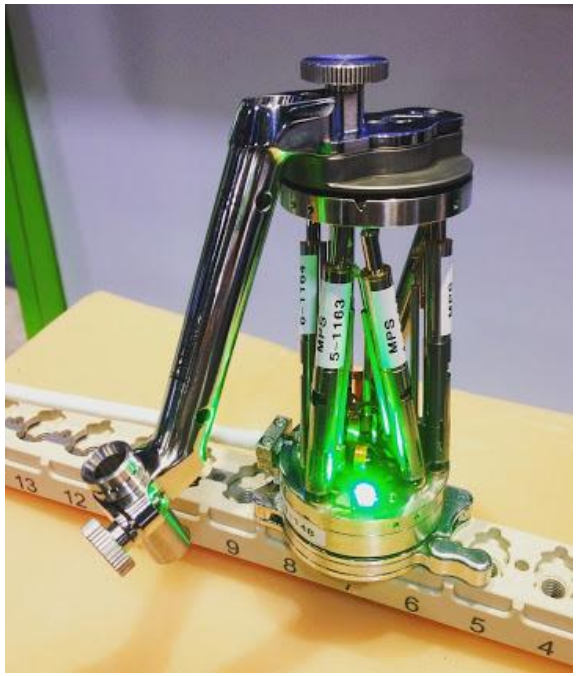
ABB Flexpicker - a delta robot based on the design by Reymond Clavel at EPFL

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

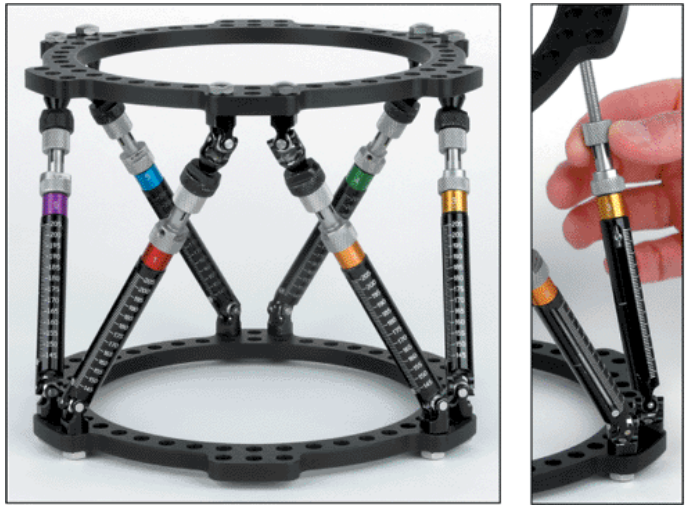
They are also popular in medical robotics



A tool for frature surgery from BRL
(120μm precision, load of 36 kg)



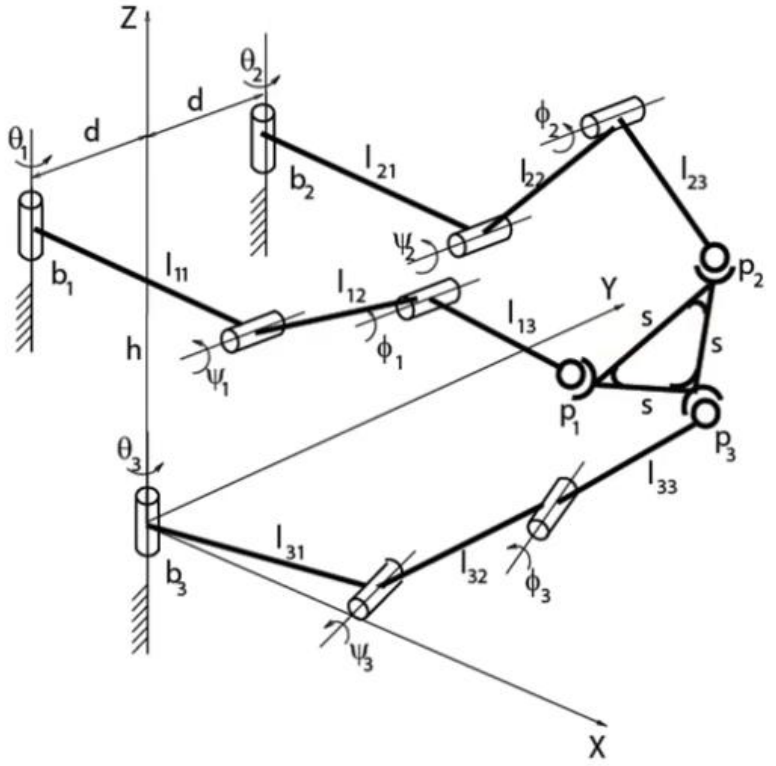
Mazor Robotics' spine surgery device



Taylor spatial frame
for fixing a and positioning limbs

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Study of parallel mechanisms can also help understand grasping with multiple fingers



Point contact with no slip can be modelled as a spherical joint

Model of a three-finger grasp

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Parallel mechanisms have high payload-to-mass ratio



DLR – based on a KUKA arm

Mass: 4.7 t
Payload: 1.3 t



BAA Training – based on a Stewart platform

Mass: 1.8 t
Payload: 10.0 t

Also, high accuracy due to structural stiffness

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

But complex kinematics and restricted workspace





TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Workspace

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Workspace

- Reachable: points that end-effector can reach – orientation can be arbitrary.
- Dexterous: point that the end-effector can reach in all its possible orientations

- Constant orientation workspace – points that can be reached in a given orientation
- Orientation workspace – all possible orientations at a given location

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

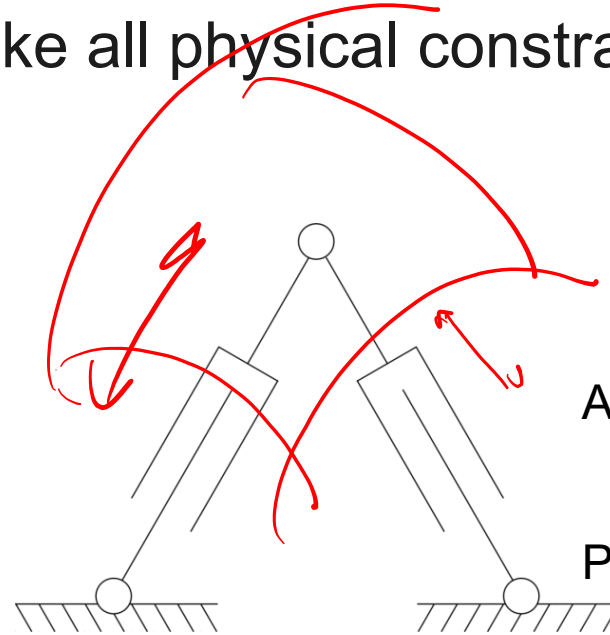
Determining the workspace by discretization

- A grid of points is formed by moving each actuator by a small amount
- Boundary of the set of nodes defines the workspace
- This is the approach you probably used in the assessment
- Disadvantage: size of increment in actuator position limits accuracy. Takes a while to compute.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

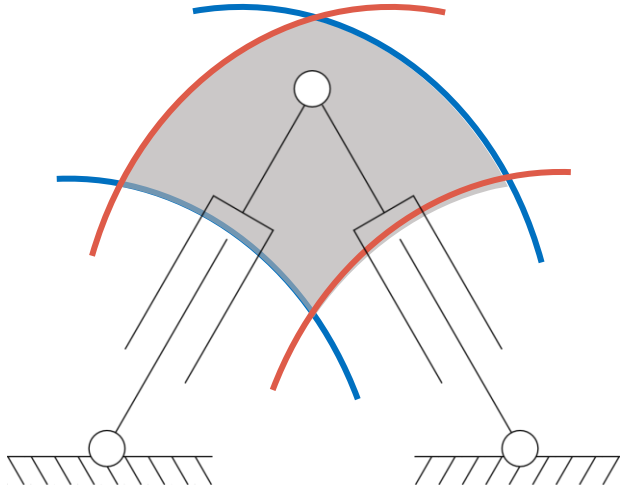
Determining the workspace from geometry

- Use minimum and maximum of each actuator's limits to define the working volume.
- Take all physical constraints into account.



Active prismatic joints

Passive revolute joints



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Mobility

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Defining mobility

- Recall the mobility equation

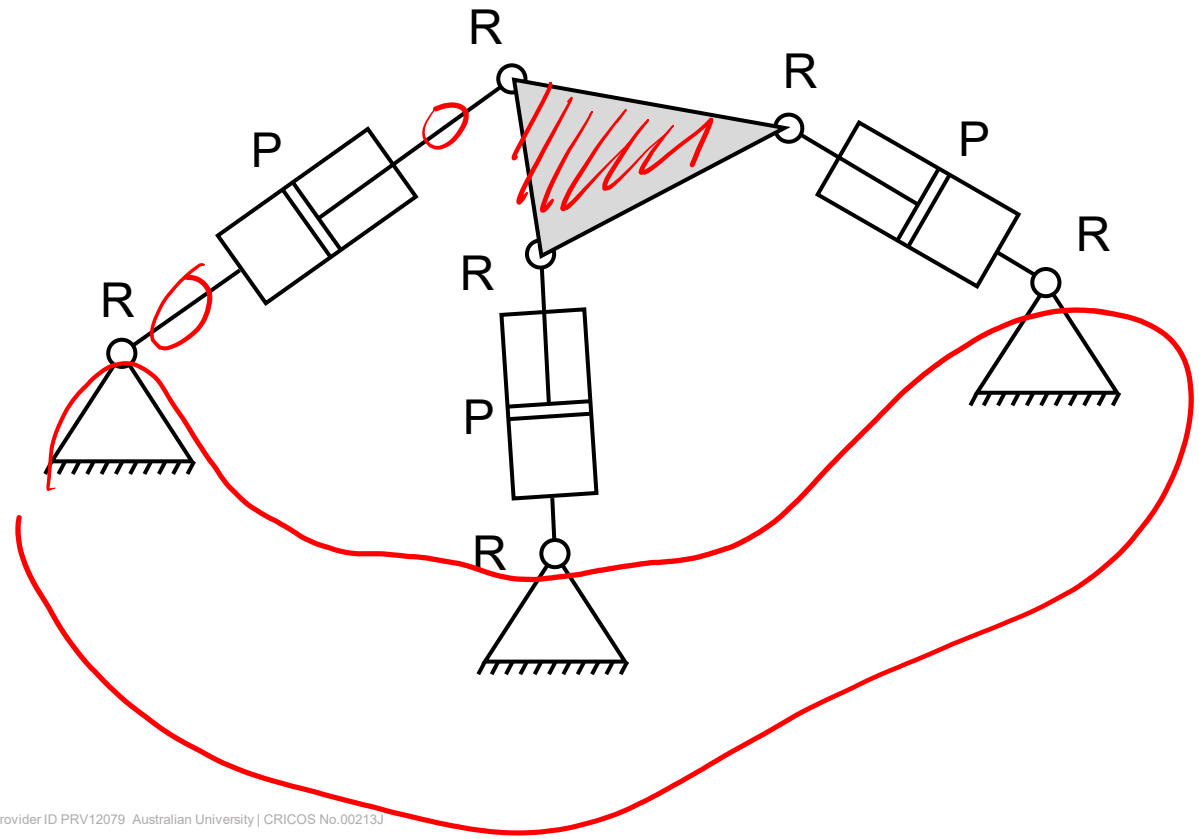
$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$

M : the mobility / DoF of the mechanism,
D : 6 for 3D and 3 for 2D
N : the number of links in the mechanism (one of which is grounded)
J : the number of joints
F_i : the degrees of freedom of joint i

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

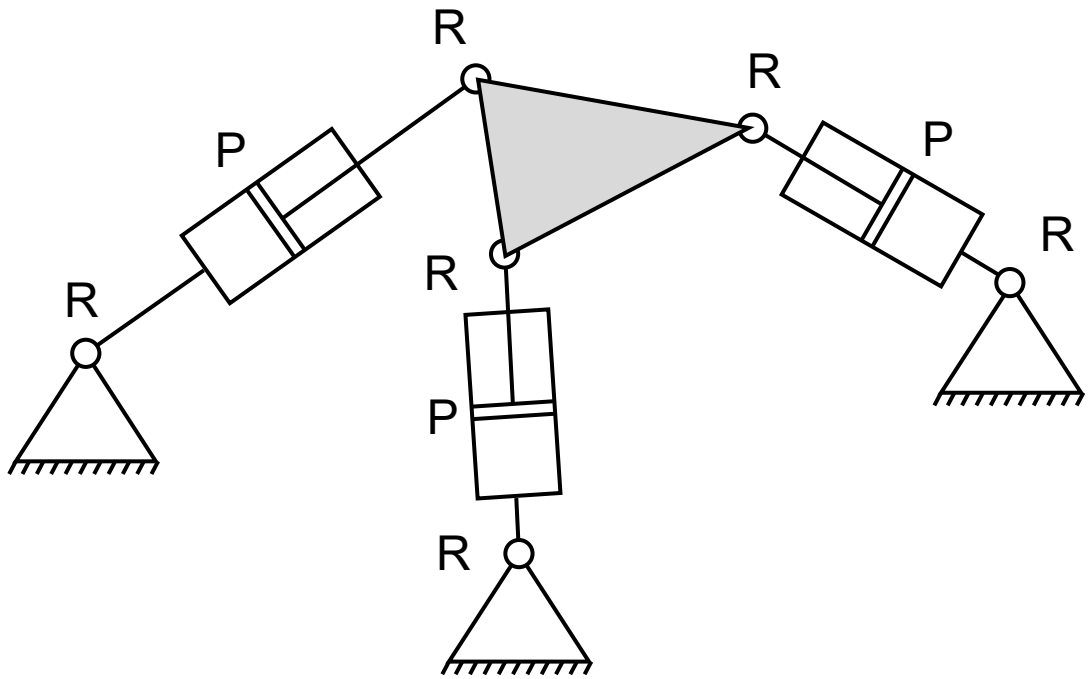
Example: Planar parallel manipulator

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Example: Planar parallel manipulator



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 3(8 - 9 - 1) + (9)$$
$$= 3$$

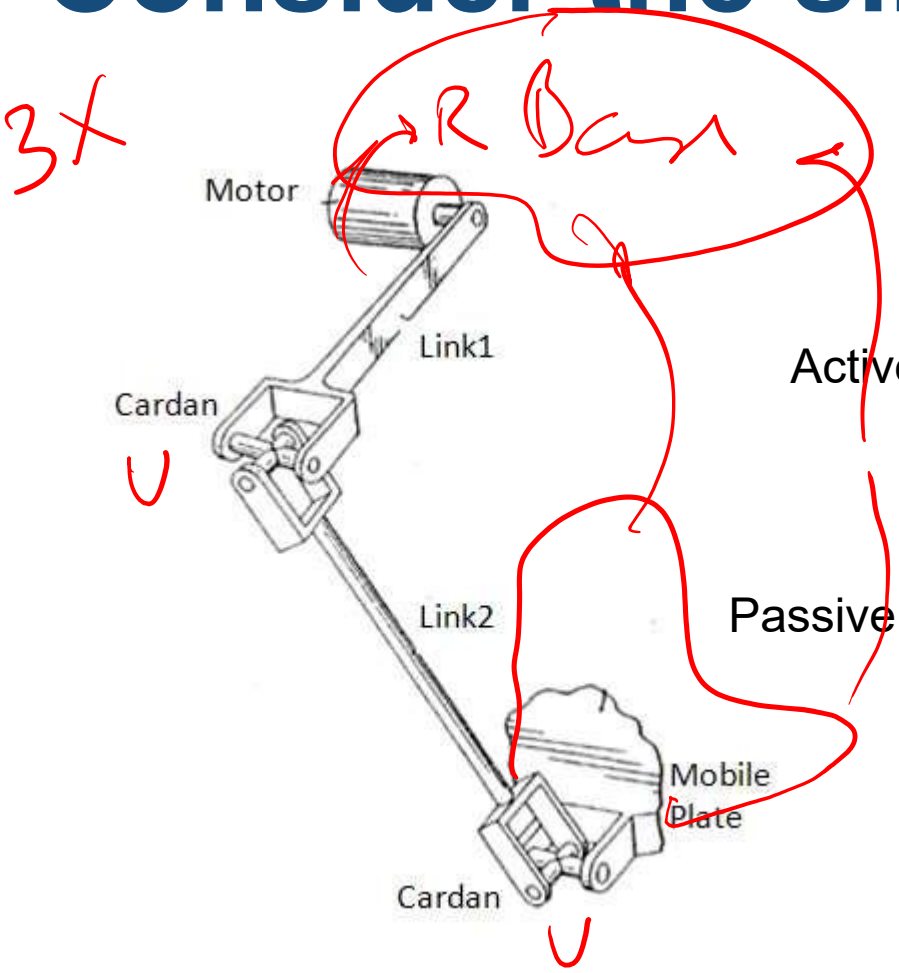
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

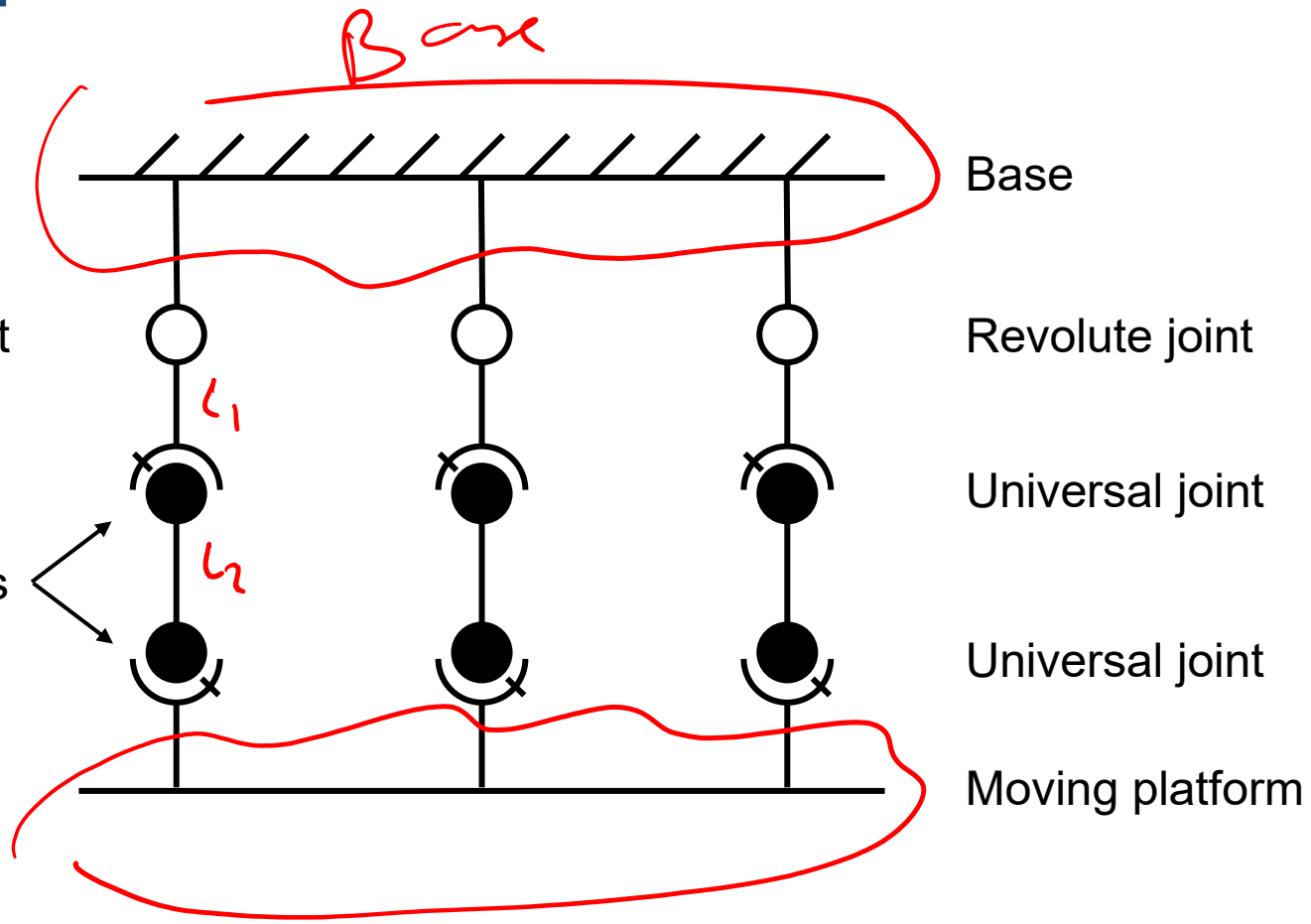
What about more complex mechanisms?



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

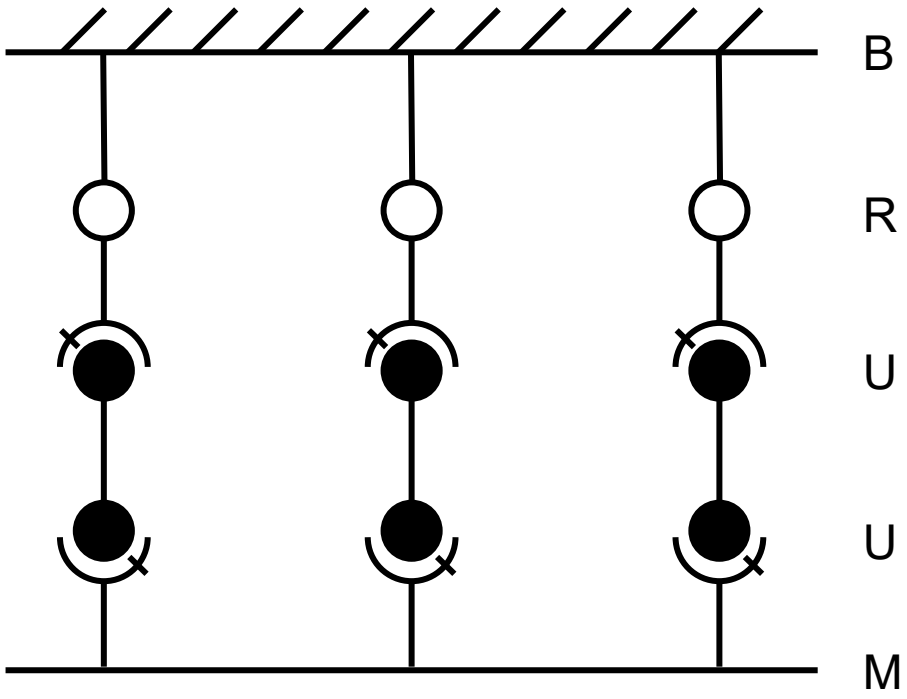
Consider the simplest case of the Delta





TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Apply the mobility equation to this mechanism



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 6(8 - 9 - 1) + (1+2+2)\times 3$$
$$= 3$$

Note: the mechanism is still 3D despite this depiction

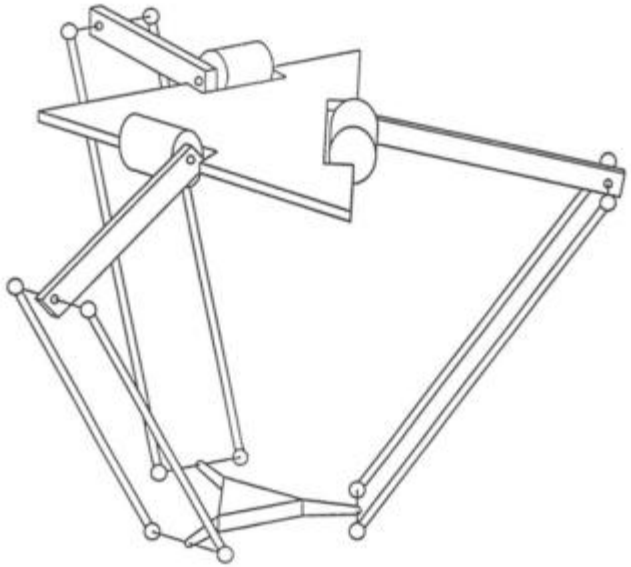
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

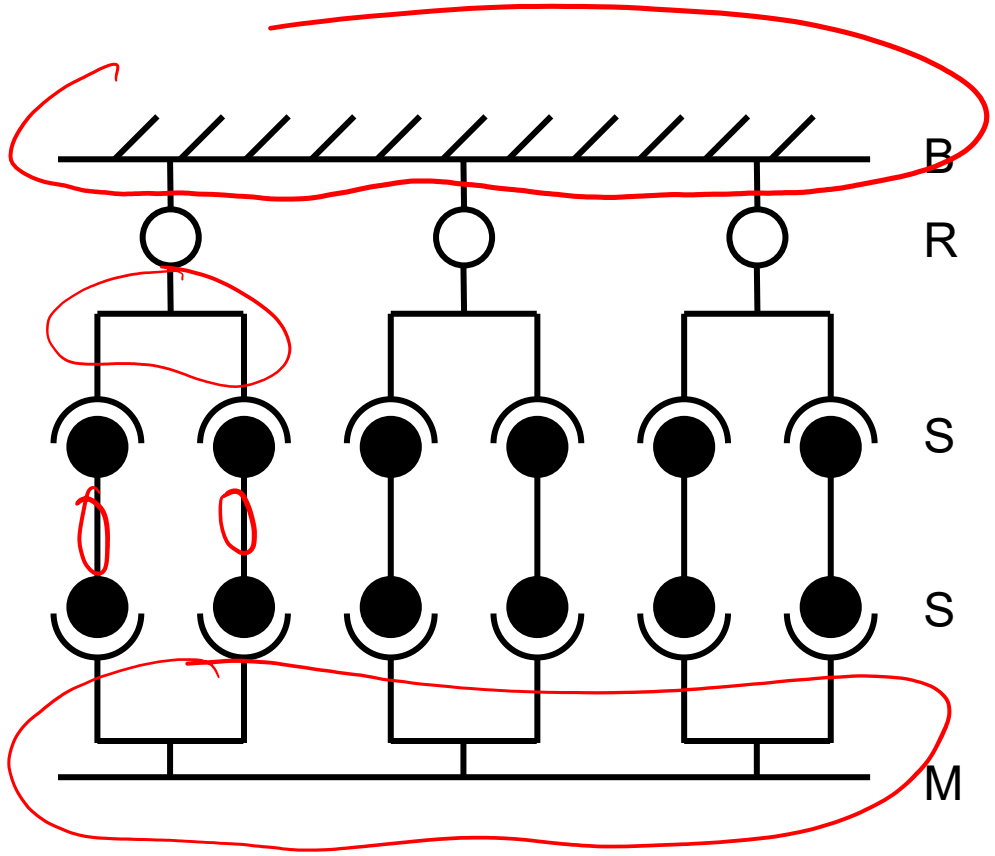
In practice, the simplest Delta is never used

- The structure is not stiff enough
- Universal joints without play are hard to find
- Play reduces frequency of operation – repeated motions cannot be performed at a fast rate

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

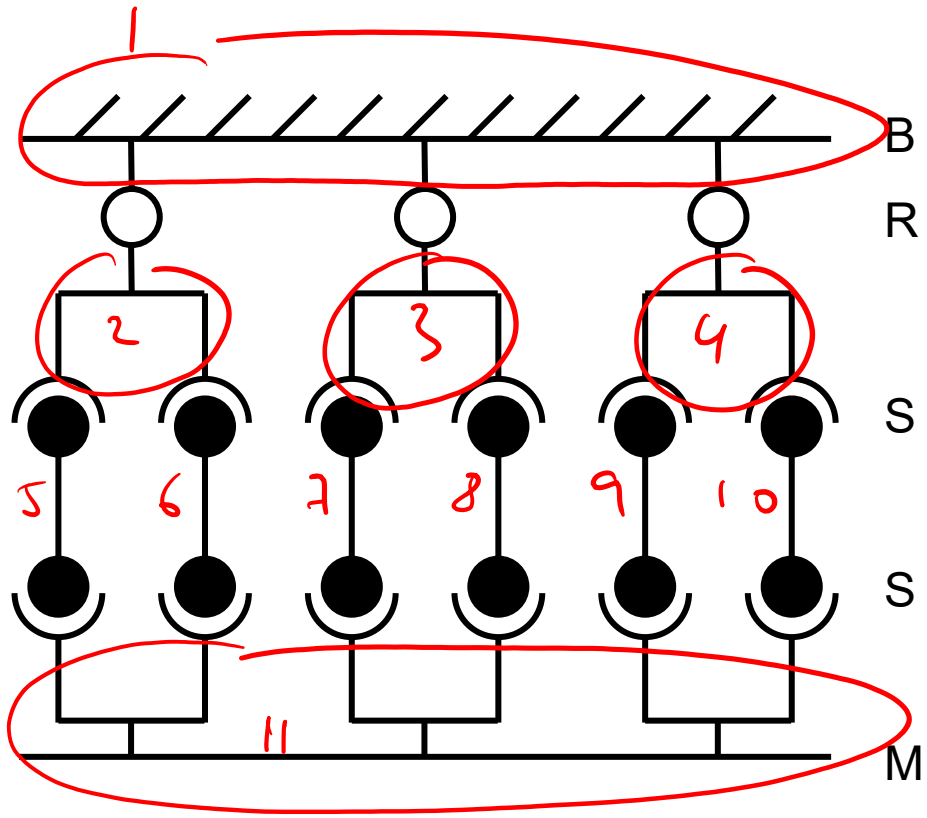
A more common configuration of the Delta





TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Apply the mobility equation to this mechanism



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$

11

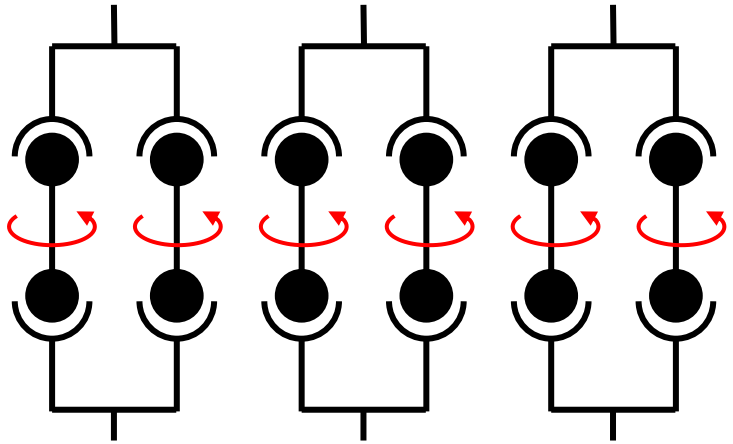
$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 6(11 - 15 - 1) + (1+4 \times 3)\times 3$$
$$= 9$$

Note: the mechanism is still 3D despite this depiction

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Are all 9 DoF influencing the mobile platform?

- The Delta robot with spherical joints and parallel bars has 6 supplementary DoF.
- They are internal to the mechanism and do not affect the motion of the mobile platform.
- They are related to the rotation of the parallel bars around their long axis.



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Extended Grübler-Kutzbach equation

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

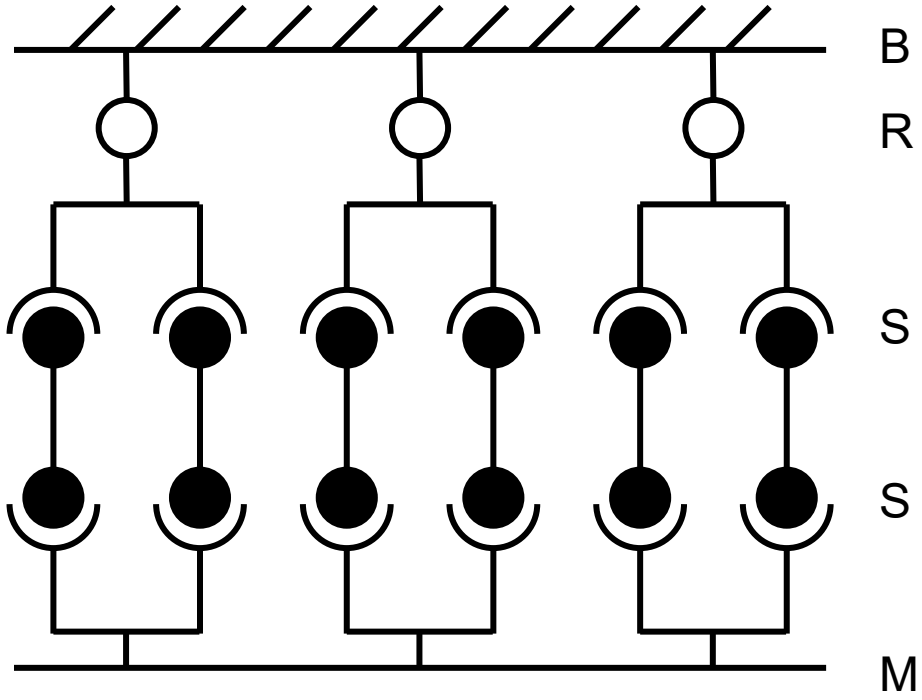
M : the mobility / DoF of the mechanism,
D : 6 for 3D and 3 for 2D
N : the number of links in the mechanism (one of which is grounded)
J : the number of joints
F_i : the degrees of freedom of joint i
R_C : number of over constraints
R_M : local mobility / DoF

$$R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$$

u_k are unique DOFs from each closed loop k

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Apply the extended mobility equation



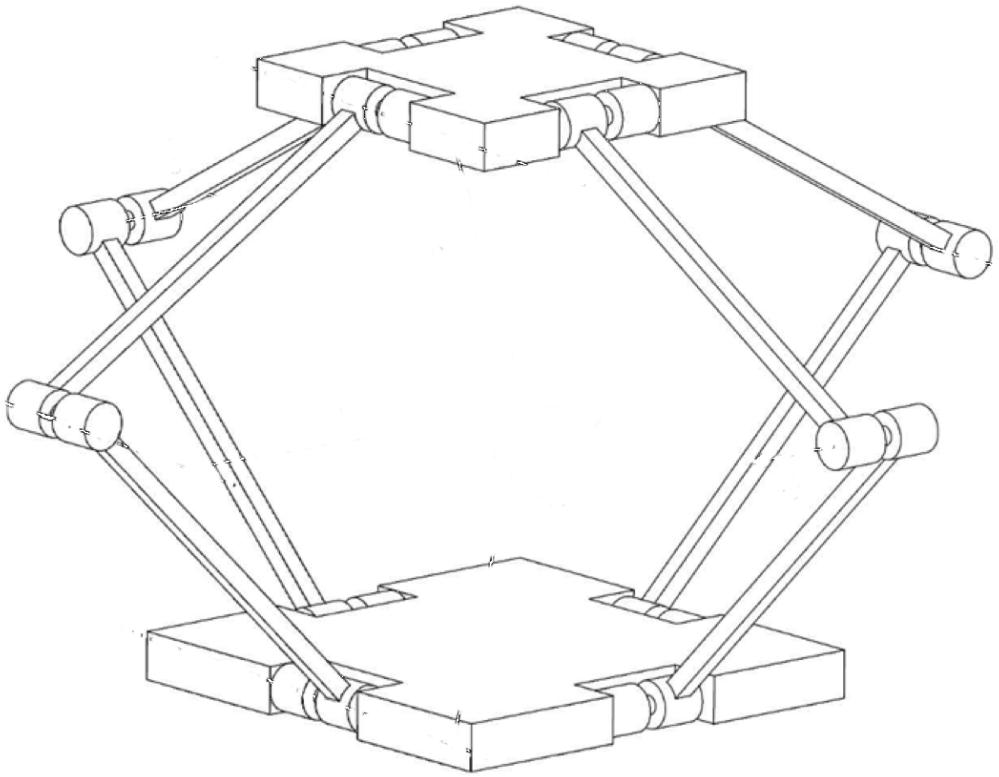
$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

$$= 6(11 - 15 - 1) + (1+4 \times 3)\times 3 + 0 - 6$$
$$= 3$$

Note: the mechanism is still 3D despite this depiction

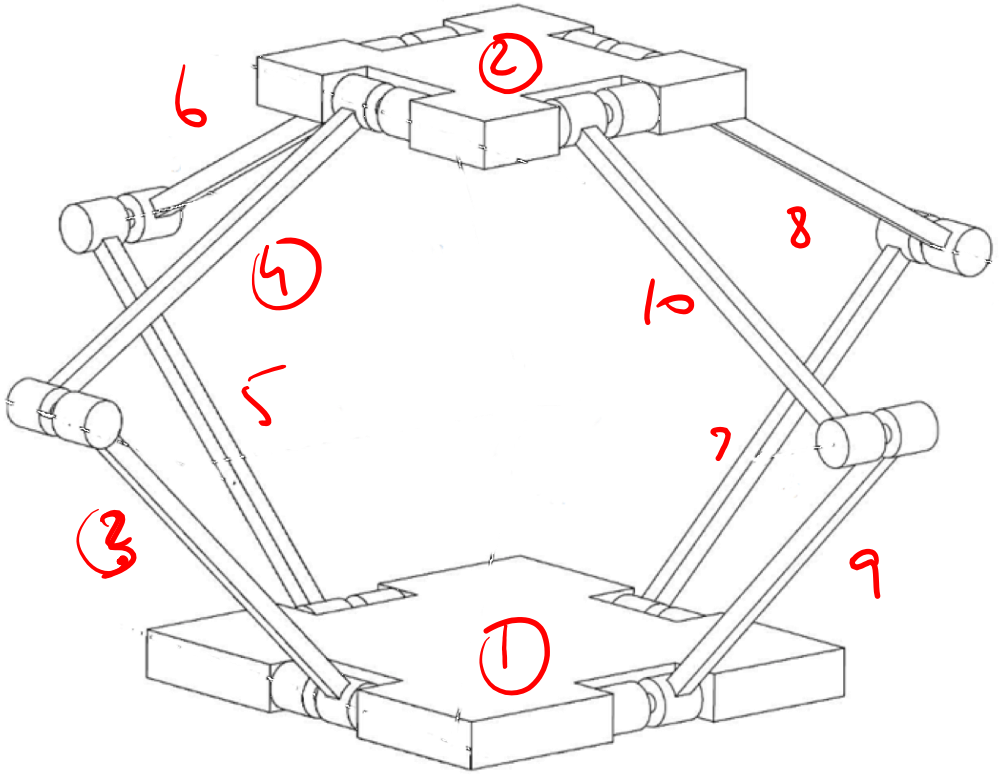
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Example: Parallel 4RRR mechanism



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Example: Parallel 4RRR mechanism



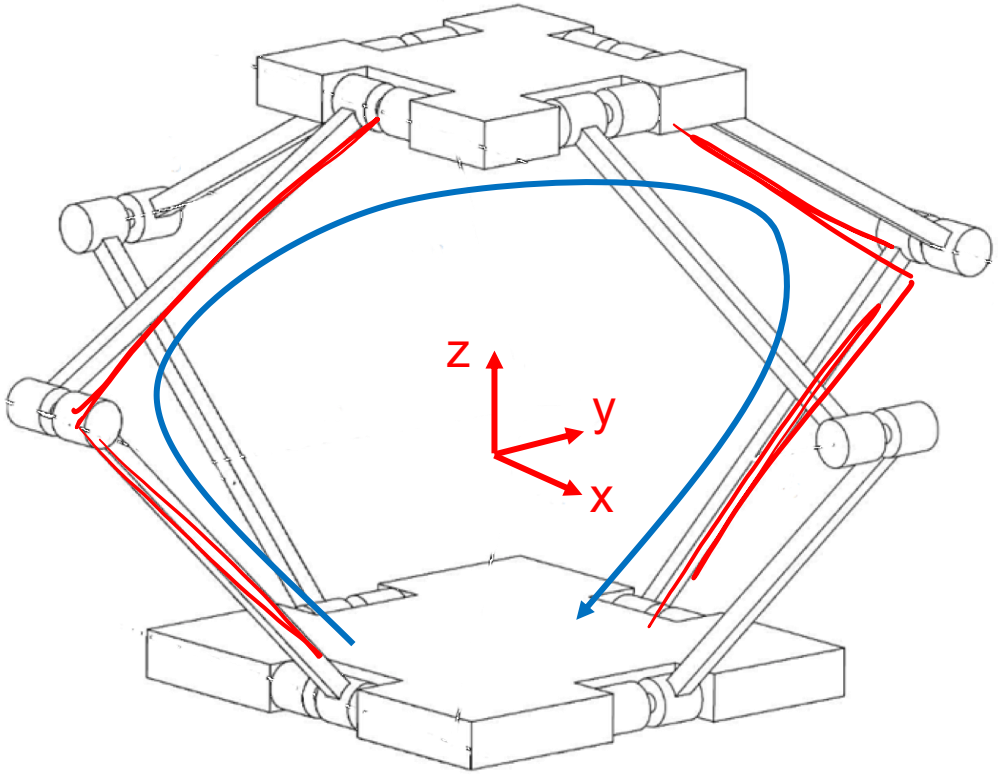
$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$
$$= 6(10 - 12 - 1) + (12) + R_C - R_M$$
$$= -6 + R_C - R_M$$

0

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Parallel 4RRR mechanism



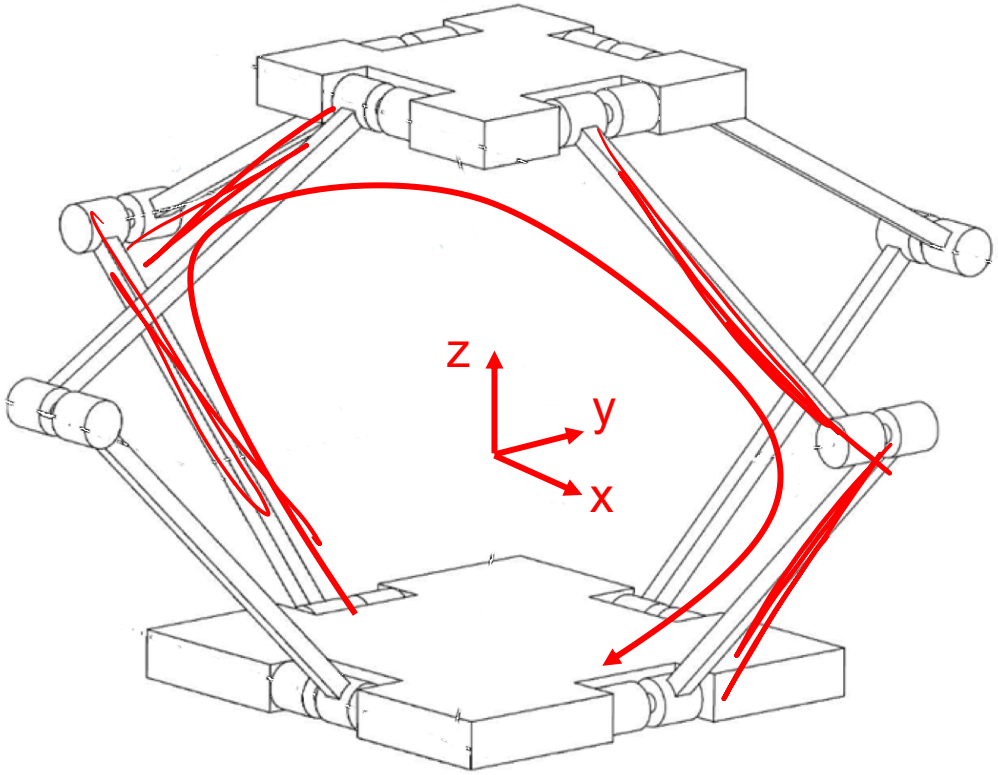
Consider this closed loop.

DoF of the moving platform due to this closed loop chain are:

T_y, T_z, R_x

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Parallel 4RRR mechanism



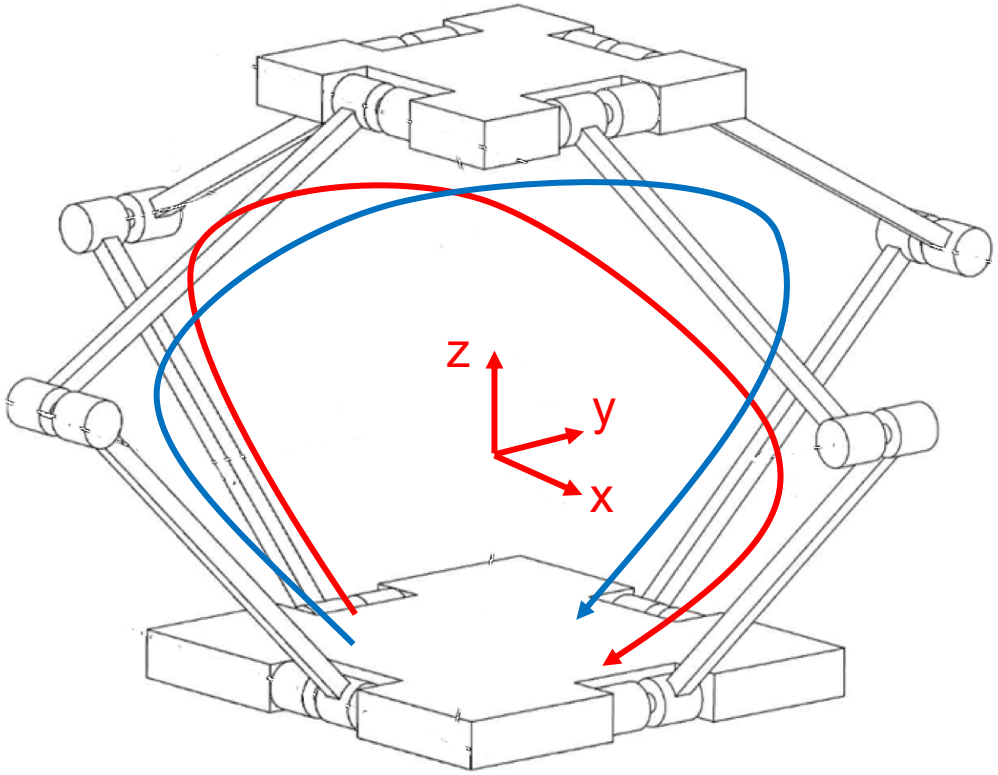
Consider this other closed loop.

DoF of the moving platform due to this closed loop chain are:

T_x, T_z, R_y

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Parallel 4RRR mechanism



Unique DoF from each loop are:

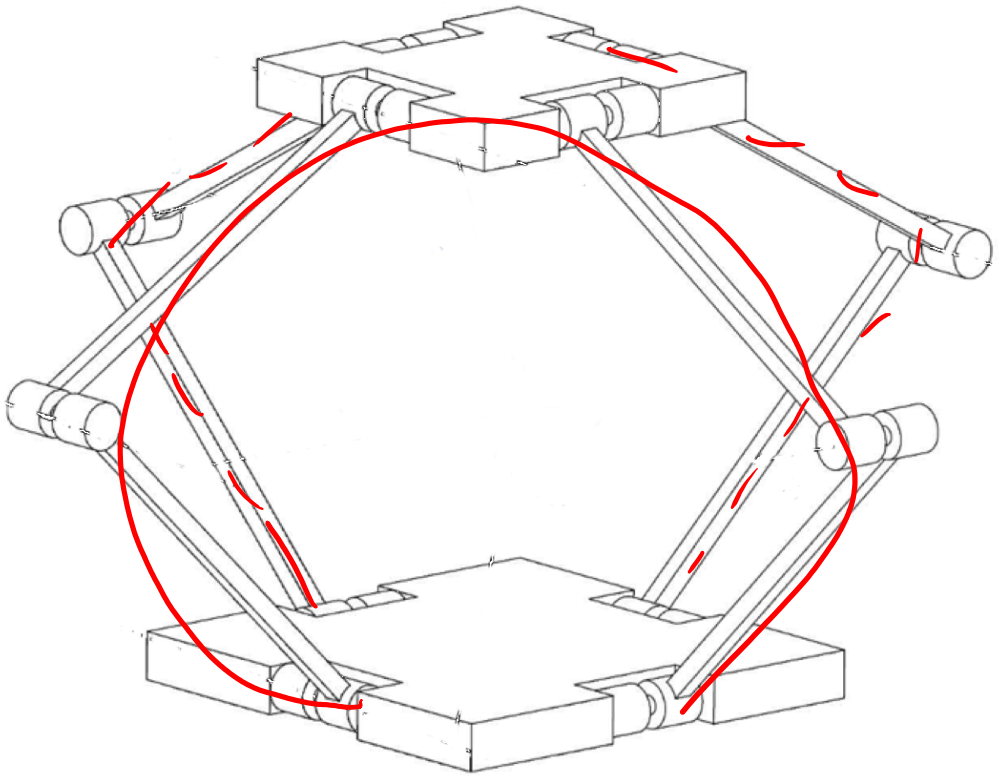
Loop 1: T_y, T_z, R_x

Loop 2: T_x, T_z, R_y

$$\sum_{k=1}^{L} u_k = 5$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Parallel 4RRR mechanism



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$
$$= 6(10 - 12 - 1) + 12 + R_C - R_M$$
$$= -6 + R_C - R_M$$

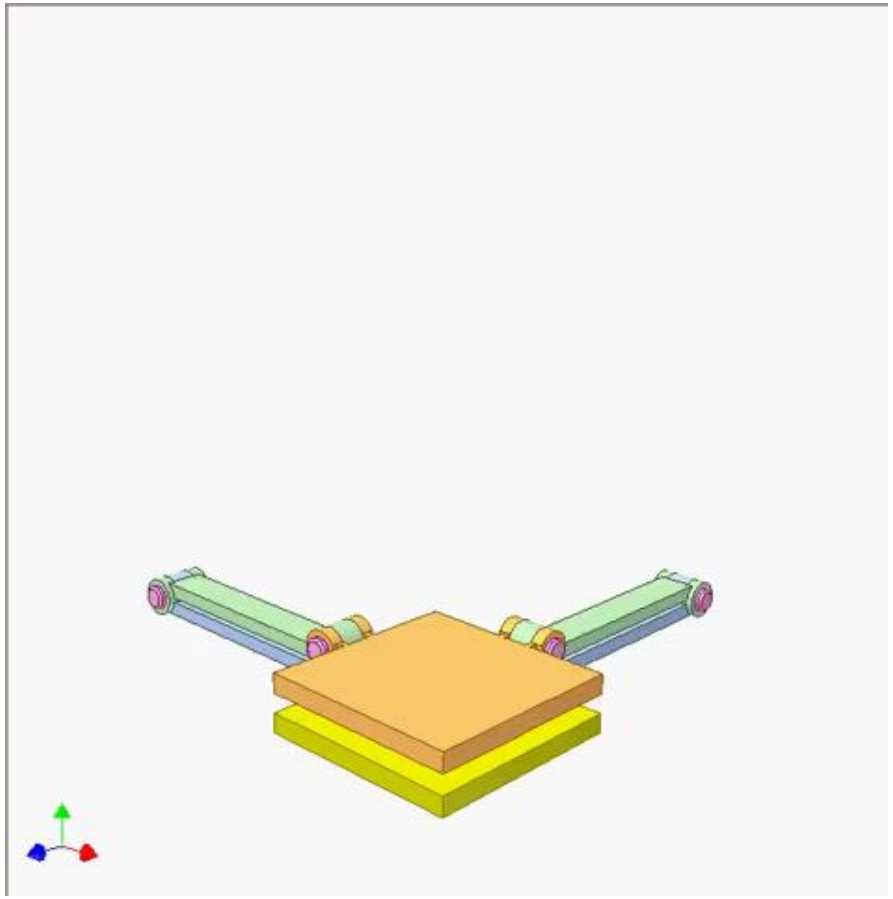
$$R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$$
$$= 12 - 5$$
$$= 7$$

$R_M = 0$

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$
$$M = -6 + 7 + 0 - 0$$
$$= 1$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

The mechanism indeed has only 1 DoF



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Break

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Configuration space

- Parallel robots typically have several more joints than the degrees of freedom
- The configuration space of a parallel robot can be split into active and passive joints
- $\mathbf{q} = \{\theta, \varphi\}$
- $\boldsymbol{\theta} \coloneqq \{\theta_i |\, \theta_i \in [0, 2\pi), i \in [1, n]\}$ is the set of n active joints
- $\boldsymbol{\varphi} \coloneqq \{\varphi_i |\, \varphi_i \in [0, 2\pi), i \in [1, m]\}$ is the set of m passive joints

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Determining passive joint coordinates using active joint coordinates

- $\mathbf{q} = \{\theta, \varphi\}$

What do we know?

- $\boldsymbol{\theta} \coloneqq \{\theta_i |\, \theta_i \in [0, 2\pi), i \in [1, n]\}$ is the set of n active joints

Can we determine these if we know the structure of the robot?

- $\boldsymbol{\varphi} \coloneqq \{\varphi_i |\, \varphi_i \in [0, 2\pi), i \in [1, m]\}$ is the set of m passive joints

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Loop closure constraint equations

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Motivation

- Given the values of n actuated joints θ_i, $i \in [1, n]$, we need a way to determine the coordinates of passive joints.
- For m passive joints, we need m independent equations to solve for φ_i, $i \in [1, m]$
- These are obtained from loop closure constraint equations

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

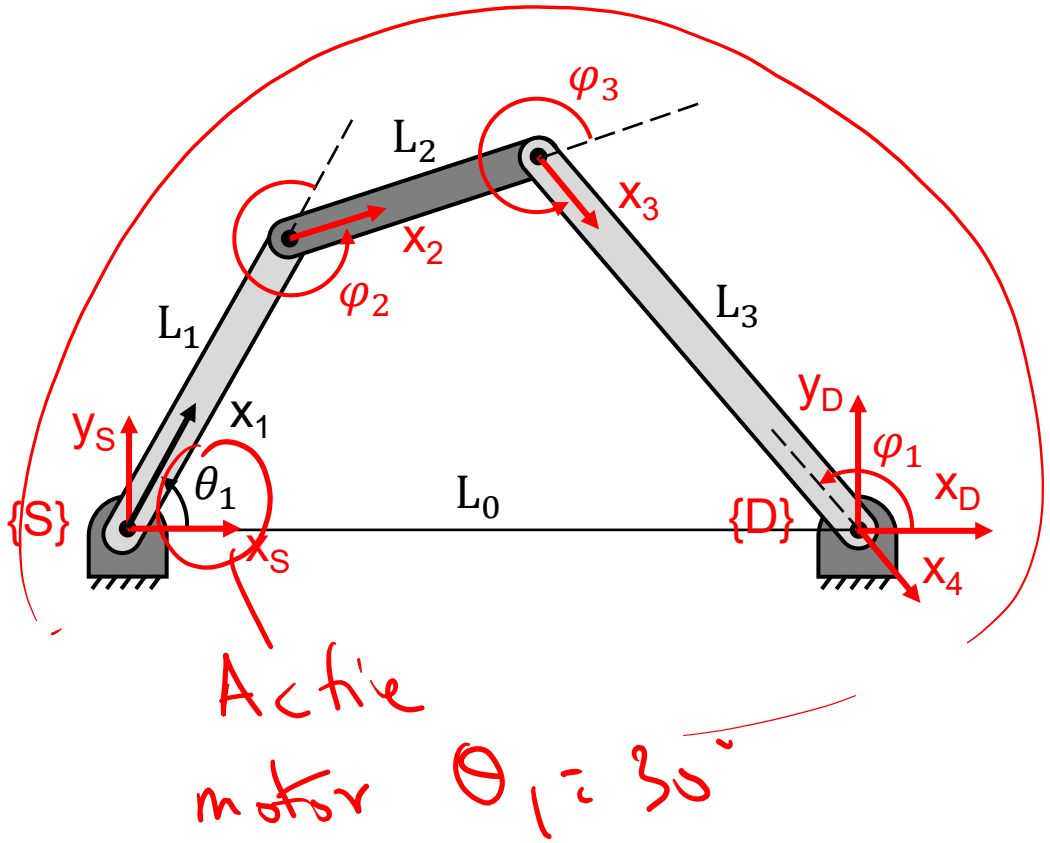
General approach to derive loop closure constrain equations

- Break the parallel manipulator into two or more serial chains
- Obtain the position and orientation of the break point for each chain
- Use joint constraint at the break point to re-join the parallel manipulator

- The trick is to break the manipulator such that
  - The number of passive variables, m is least, and
  - Minimum number of constraint equations $\eta_i(q) = 0$, $i \in [1, m]$ are used

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

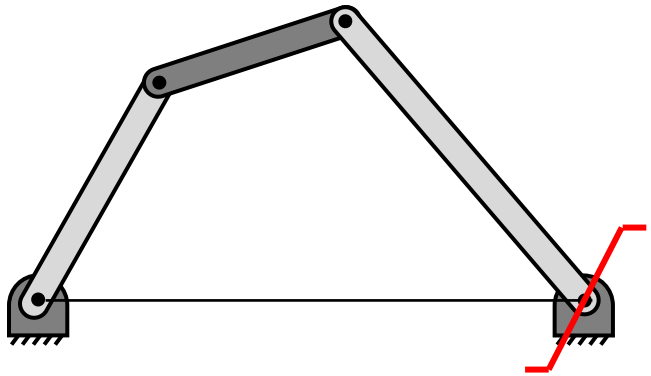
Example: planar 4 bar mechanism

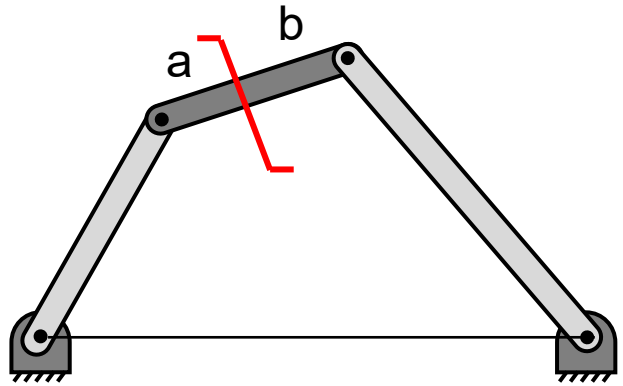


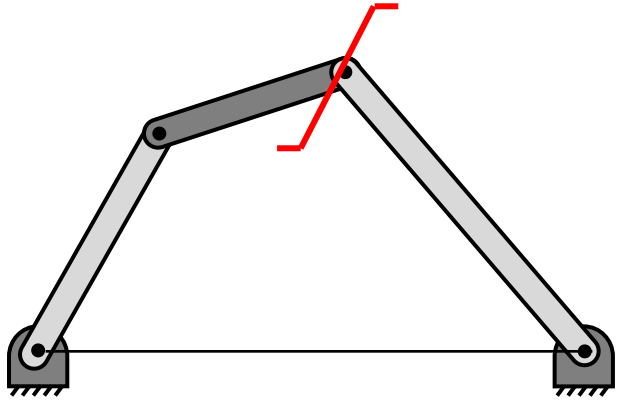
- Contains one loop
- Relevant frames are {S}, {1}, {2}, {3}, {4} and {D}
- Say θ_1 is the active joint
- φ_1, φ_2 and φ_3 are passive joints

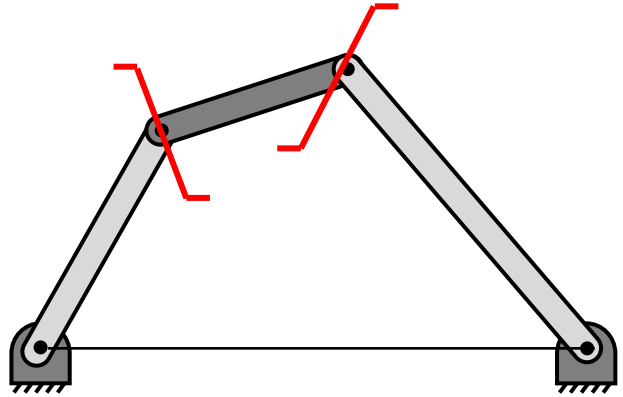
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

There are many ways to break the loop



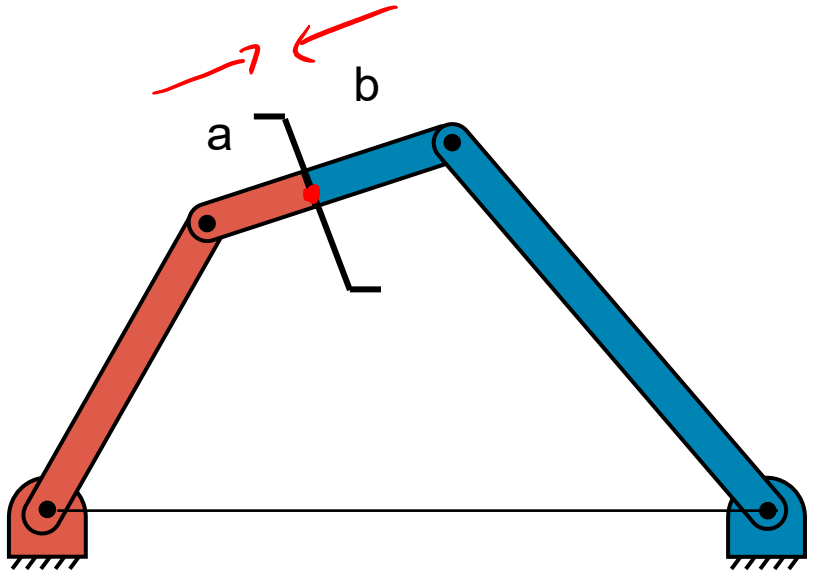






TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

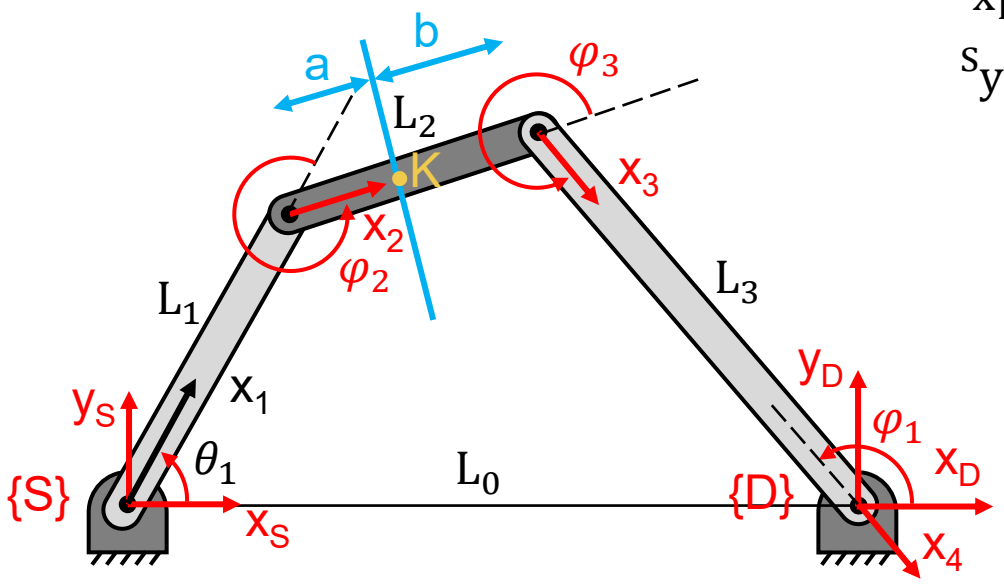
Consider the case where we break link 2



This gives two RR type serial link manipulators

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Let us write the coordinate of the break point



$${}^{S}x_K =$$

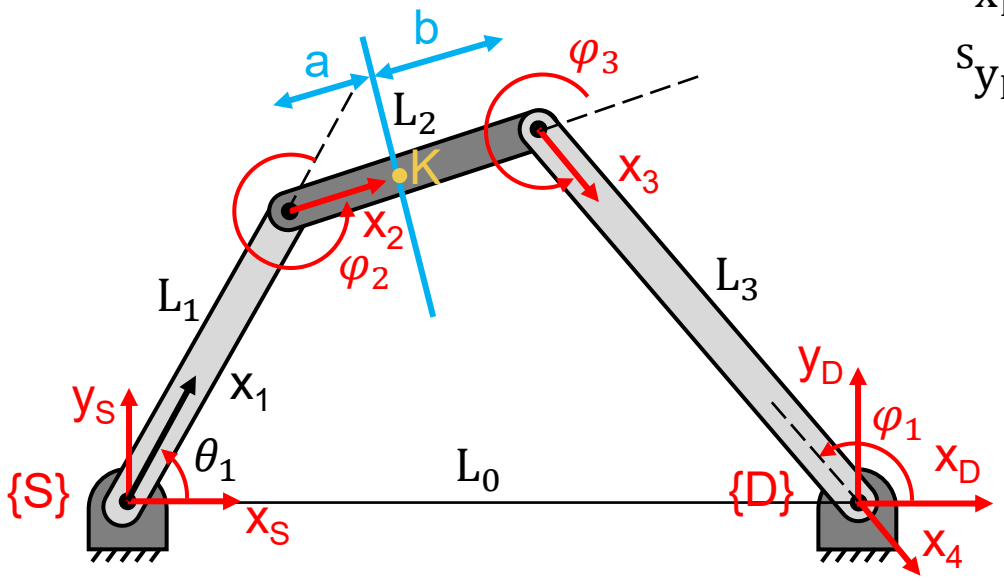
$${}^{S}y_K =$$

$${}^{D}x_K =$$

$${}^{D}y_K =$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Let us write the coordinate of the break point



$$^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2)$$
$$^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2)$$

$$^{D}x_K = L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$
$$^{D}y_K = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

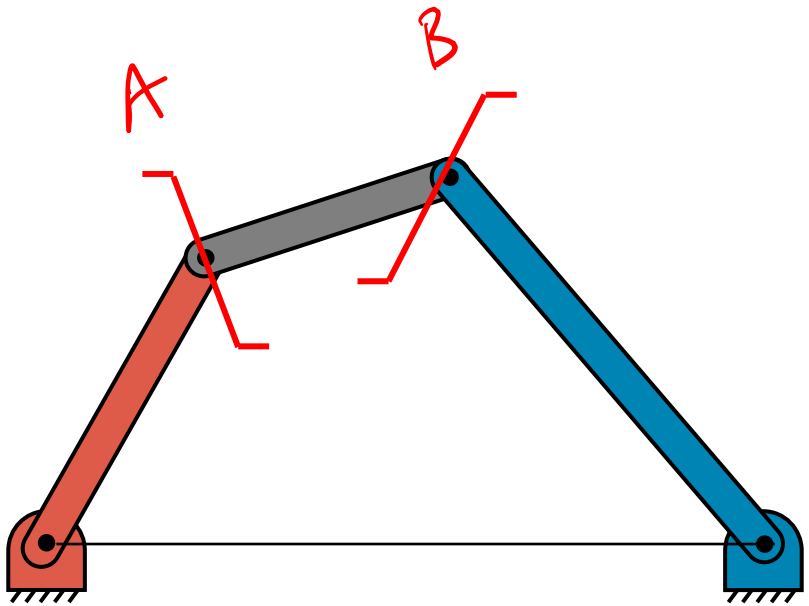
Impose the constraint that the broken link is rigid

$$^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$
$$^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2) = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$
$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

We now have 3 independent equations from which we can solve for φ_1, φ_2 and φ_3.
$n = 1$, $m = 3$, $J = 4$

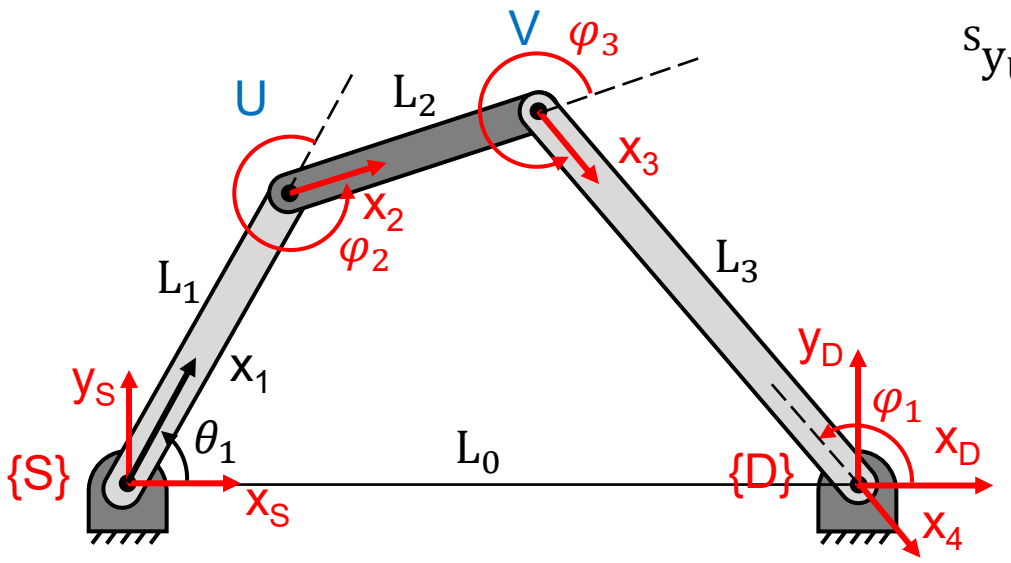
TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Now consider the case where we break on either side of link 2



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Let us write the coordinates of the break points



$${}^{S}x_U =$$

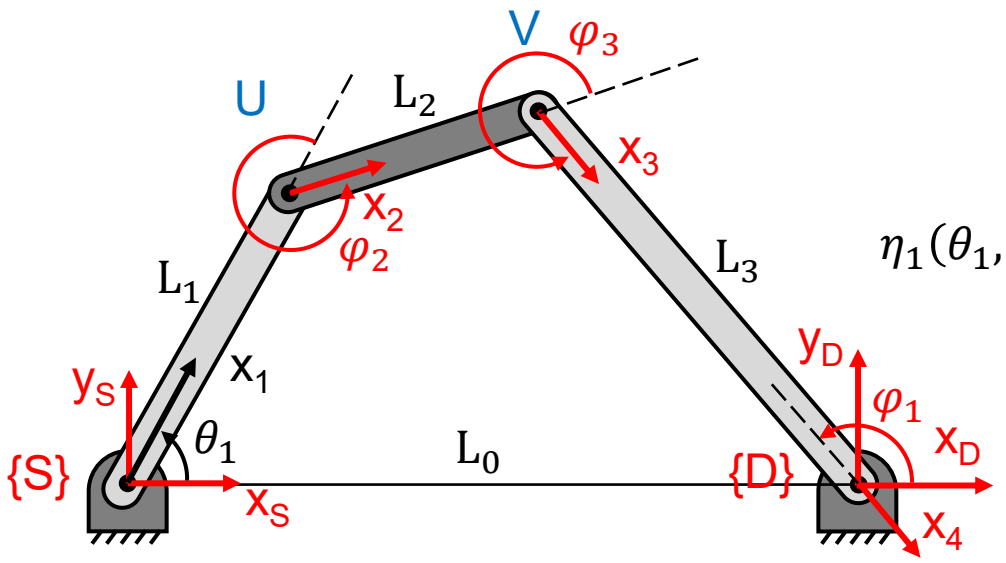
$${}^{S}y_U =$$

$${}^{D}x_V =$$

$${}^{D}y_V =$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Let us write the coordinates of the break points



$$^{S}x_U = L_1 \cos\theta_1 \qquad ^{D}x_V = L_3 \cos\varphi_1$$

$$^{S}y_U = L_1 \sin\theta_1 \qquad ^{D}y_V = L_3 \sin\varphi_1$$

Impose the constraint that the length of the link is L_2

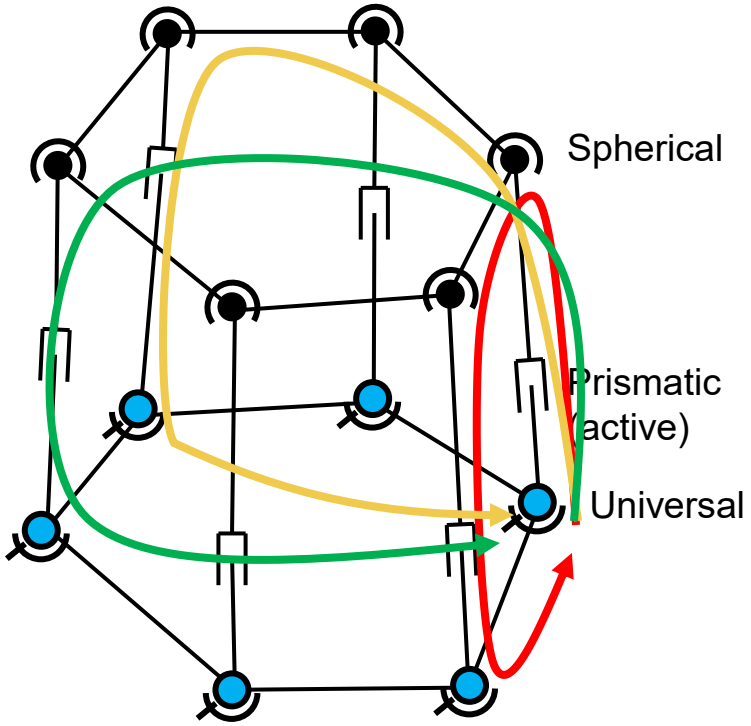
$$\eta_1(\theta_1, \varphi_1) = (L_1 \cos\theta_1 - L_0 - L_3 \cos\varphi_1)^2 + (L_1 \sin\theta_1 - L_3 \sin\varphi_1)^2 - L_2^2 = 0$$

We have only one constraint equation. $n = 1$, $m = 1$, $J = 4$

The above equation is called the Freudenstein equation, in the study of four-bar mechanisms

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

What about the Gough-Stewart platform?



- 6 active and 30 passive parameters
- Many loops are possible
- Each loop potentially has 6 independent equations!
- Deciding which 30 of them to choose is not trivial

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Forward Kinematics

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

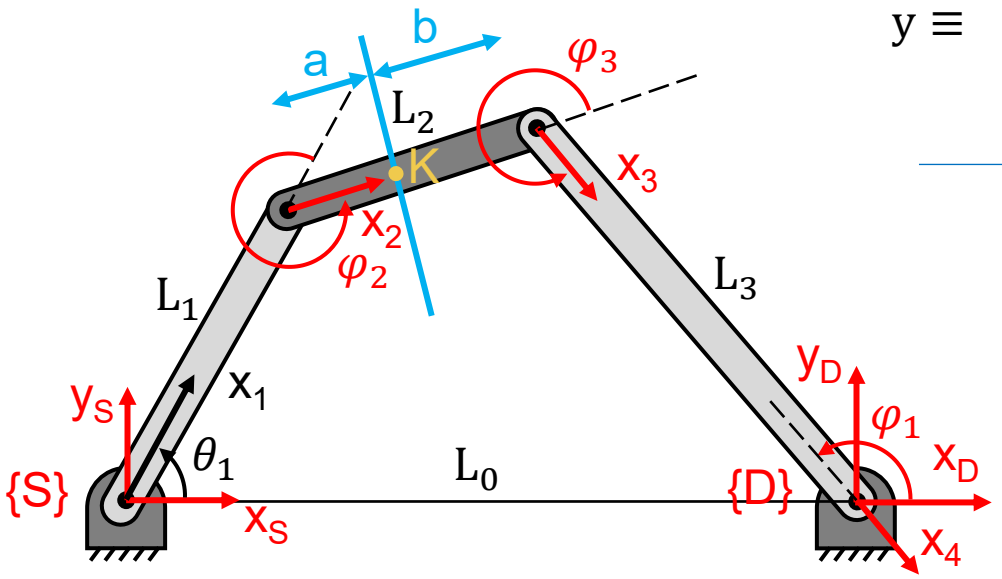
The problem

- Given: the geometry of the robot and position of the active joints θ_i, $i \in [1, n]$
- Determine: position and orientation of the end effector
- Step 1: Obtain the m passive joint variables
  - Obtain m loop closure constraint equations in m passive and n active joint variables.
  - Solve the m non-linear equations, if possible, to obtain the passive joint variables φ_i, $i \in [1, m]$
- Step 2: Obtain the position and orientation of the output link from θ_i and φ_i
- There is no general method like in the case of serial manipulators

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Solvable Example: four bar mechanism

Consider the four-bar mechanism in which we broke link 2 at a point.



Loop closure constraints gave us

$$x \equiv {}^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$

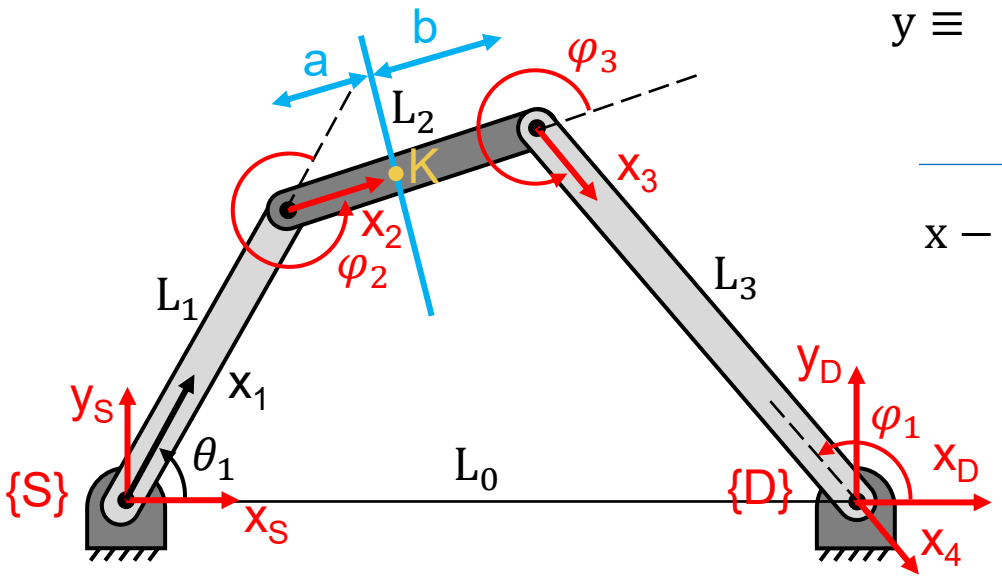
$$y \equiv {}^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2) = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Solvable Example: four bar mechanism

Consider the four-bar mechanism in which we broke link 2 at a point.



Loop closure constraints gave us

$$x \equiv {}^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$

$$y \equiv {}^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2) = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

$$x - L_0 = L_3 \cos\varphi_1 - b\cos(\theta_1 + \varphi_2) \qquad y = L_3 \sin\varphi_1 - b\sin(\theta_1 + \varphi_2)$$

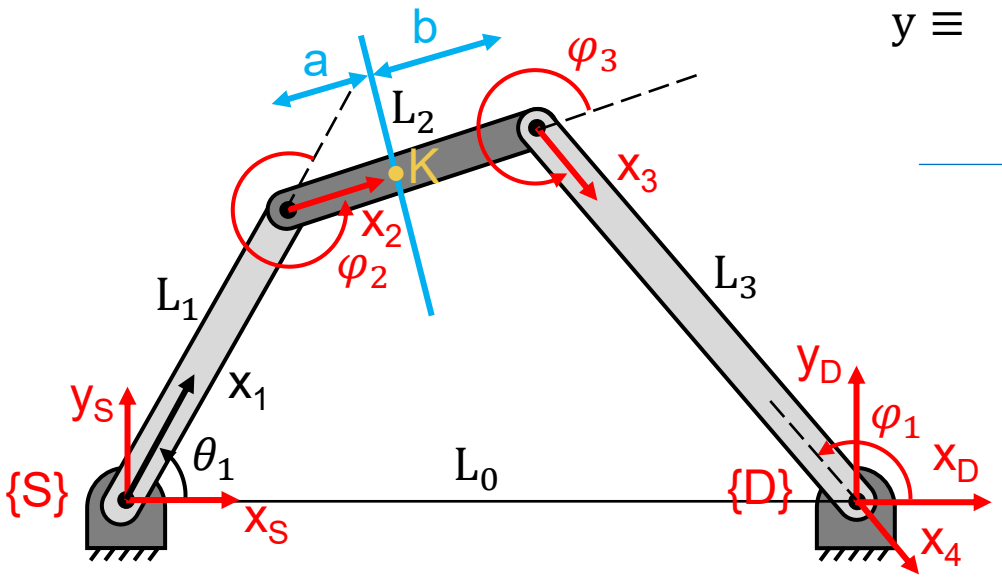
Let $\delta = \theta_1 + \varphi_2$. Squaring and adding,

$$A_1 \cos\delta + B_1 \sin\delta + C_1 = 0$$

$$A_1 = x - L_0,\ B_1 = y,\ C_1 = \left(\frac{1}{2b}\right)[(x - L_0)^2 + (y)^2 + (b)^2 - (L_3)^2]$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

FK of four-bar mechanism – continued



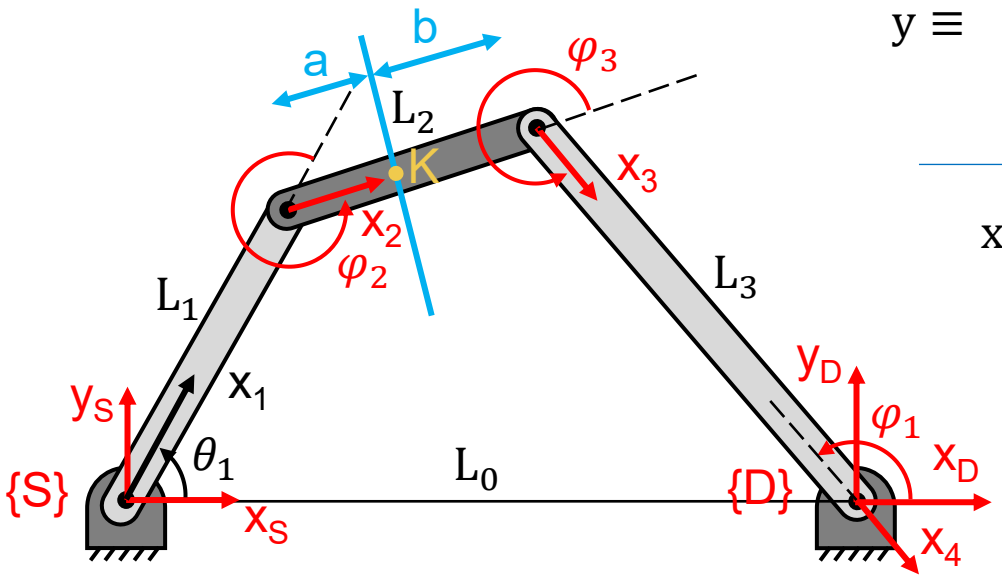
$$x \equiv {}^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$

$$y \equiv {}^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2) = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

FK of four-bar mechanism – continued



$$x \equiv {}^{S}x_K = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$

$$y \equiv {}^{S}y_K = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2) = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

$$\theta_1 + \varphi_2 = \varphi_1 + \varphi_3 + \pi$$

$$x = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2) \qquad y = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2)$$

Again, let $\delta = \theta_1 + \varphi_2$. Squaring and adding,

$$A_2 \cos\delta + B_2 \sin\delta + C_2 = 0$$

$$A_2 = x,\ B_2 = y,\ C_2 = \left(\frac{1}{2a}\right)[(L_1)^2 - (a)^2 - (x)^2 - (y)^2]$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Solve the two equations that we obtained

$$A_1 \cos\delta + B_1 \sin\delta + C_1 = 0 \qquad A_2 \cos\delta + B_2 \sin\delta + C_2 = 0$$

Use tangent half angle substitution and solve

$$\delta = \tan^{-1}\left(\frac{A_1C_2 - A_2C_1}{(B_1C_2 - B_2C_1) + (A_1B_2 - A_2B_1)}\right) \qquad (A_1B_2 - A_2B_1)^2 = (A_1C_2 - A_2C_1)^2 + (B_1C_2 - B_2C_1)^2$$

Simplifies to

$$4a^2b^2{L_0}^2y^2 = \left[b(x - L_0)\left({L_1}^2 - a^2 - x^2 - y^2\right) - ax\left((x - L_0)^2 + y^2 + b^2 - {L_3}^2\right)\right]^2 + y^2\left[b\left({L_1}^2 - a^2 - x^2 - y^2\right) - a\left((x - L_0)^2 + y^2 + b^2 - {L_3}^2\right)\right]^2$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Obtain other passive joint parameters

- θ_1 is given
- $\varphi_2 = \delta - \theta_1$
- φ_1 can be obtained from Freudenstein's equation

$$\eta_1(\theta_1, \varphi_1) = (\mathrm{L}_1 \cos\theta_1 - \mathrm{L}_0 - \mathrm{L}_3 \cos\varphi_1)^2 + (\mathrm{L}_1 \sin\theta_1 - \mathrm{L}_3 \sin\varphi_1)^2 - \mathrm{L}_2{}^2 = 0$$

- $\varphi_3 = \theta_1 + \varphi_2 - \varphi_1 - \pi$

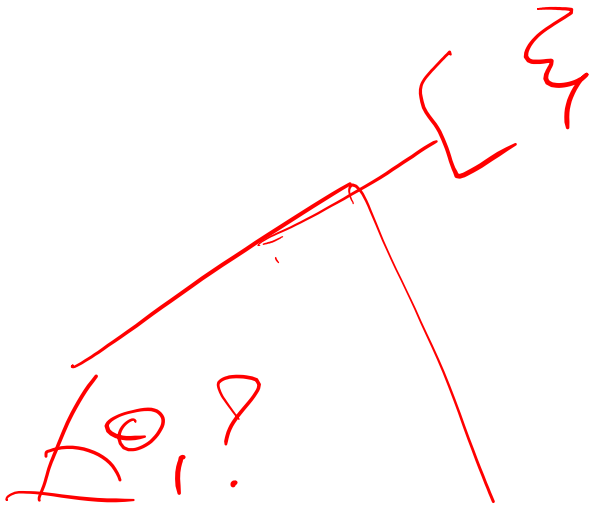
Now choose your end effector and determine its position and orientation

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

What about more complex mechanisms?

- Multiple solutions may exist
- Some solutions may have imaginary values for the passive joint parameters
- Numerical approaches may also be used

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J



Inverse Kinematics

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

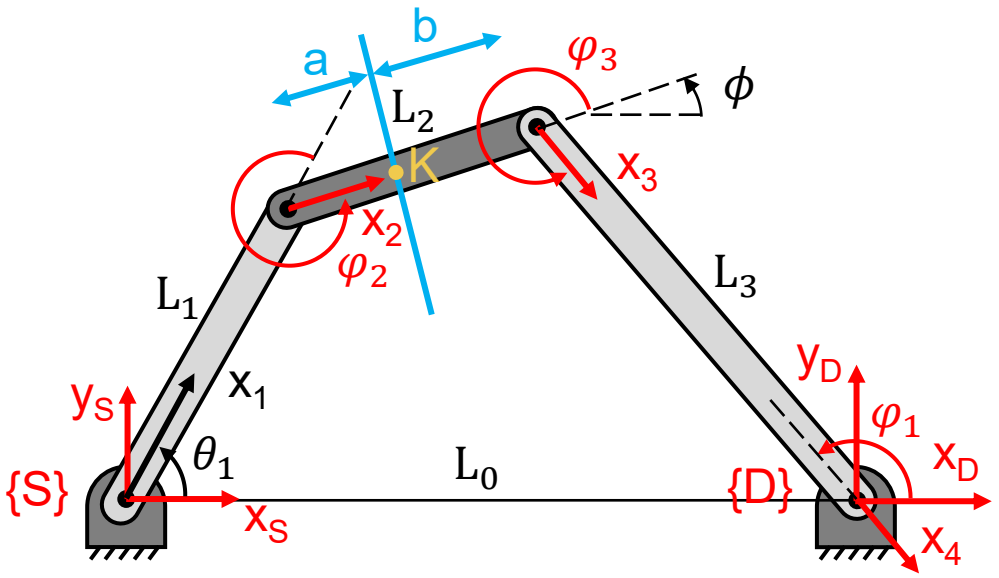
The inverse kinematics problem

- Given: the geometry of the robot and the pose of a chosen output link.
- Determine: the active and passive joint variables.
- Simpler than FK. No need to consider loop closure constraint equations.
- General idea:
  - break the mechanism into serial chains
  - obtain the joint angles of each chain.

- Note: make sure that the serial chain is not redundant – no unique solution.

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

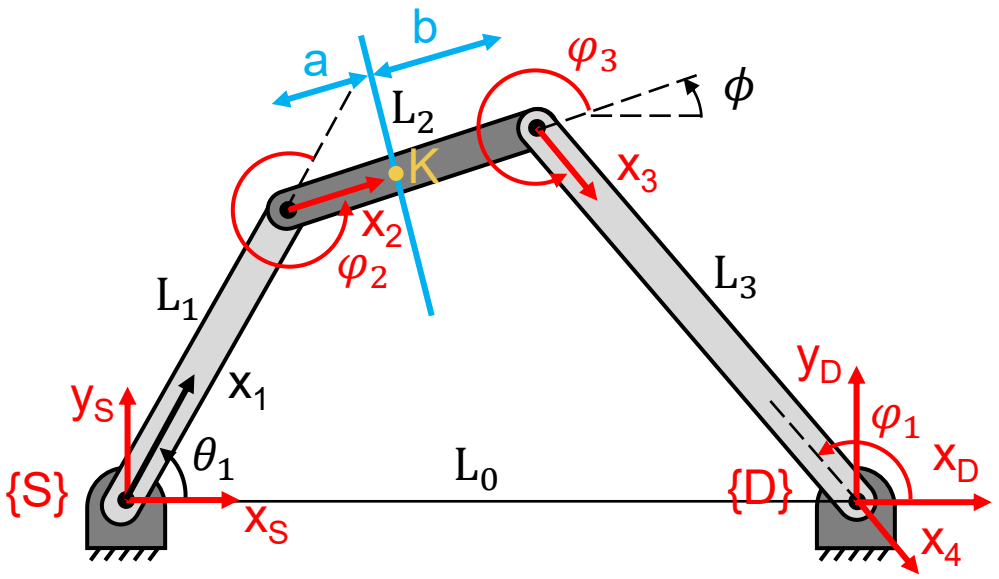
Example: four bar mechanism



- Given:
  - The position of a point K on the output link
  - The orientation ϕ of the output link (alternatively, ${}^{S}R_2$ of link 2 is given)
- Known:
  - L_0, L_1, L_2, L_3
  - $L_2 = a + b$
  - a and b are known
- Determine:
  - θ_1, φ_1, φ_2 and φ_3

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Start from what is given



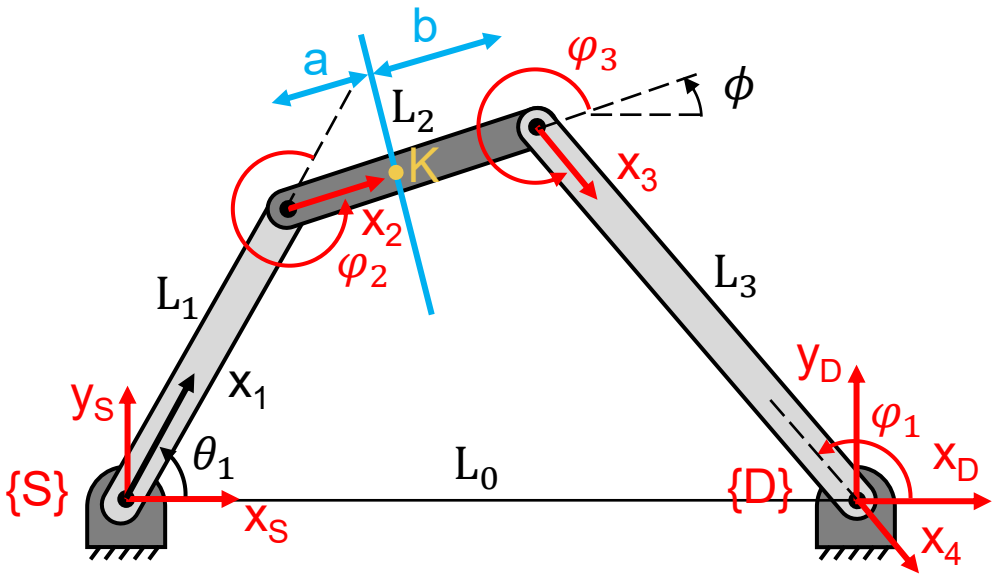
$$x = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2)$$
$$y = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2)$$
$$\phi = \theta_1 + \varphi_2 - 2\pi$$

Solve for θ_1 and φ_2

$$\theta_1 = \text{atan2}(y - a\sin\phi\,, x - a\cos\phi)$$
$$\varphi_2 = \phi - \theta_1 + \cancel{2\pi}$$

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Four bar IK - continued



$$x = L_0 + L_3 \cos\varphi_1 + b\cos(\varphi_1 + \varphi_3)$$

$$y = L_3 \sin\varphi_1 + b\sin(\varphi_1 + \varphi_3)$$

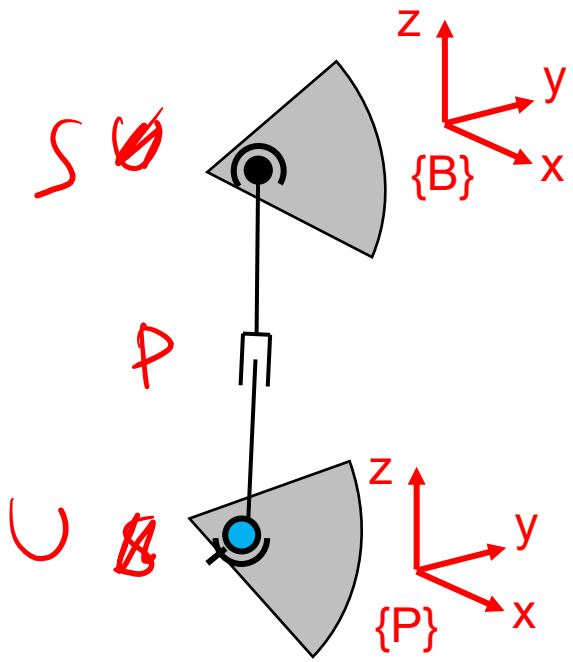
$$\phi = \varphi_1 + \varphi_3 - \pi$$

Solve for φ_1 and φ_3

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Homework: IK of the Gough-Stewart platform

- The leg has one active prismatic joint
- The spherical and universal joints are passive
- You are given ${}^{B}T_{P}$ – the position and orientation of the movable platform P wrt B
- Determine all joint angles



TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Comparison between Serial and Parallel mechanisms

	Serial	Parallel
Type of kinematic chain	Open chains	Closed chains
Joint types	Active	Active and Passive
FK problem	Simple	Complex
IK problem	Complex	Simple
Joint errors	Cumulative	Non-cumulative
Work volume	Large	Small
Payload to weight ratio	Low	Very high
Inertia of moving parts	High	Low

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Summary

- Parallel robots have benefits over serial robots – large payloads, more accurate
- Parallel robots typically have more joints than DoF
- Forward kinematics problem is not easy to solve
- Inverse kinematics is an easier problem to solve

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J

Next week

- Velocity kinematics

TEQSA Provider ID PRV12079 Australian University | CRICOS No.00213J